Quiz 1 – Calculus I (Class 64)

1. Solve the inequality $3 \leq |2x - 5| \leq 7$. Express the solution as a *set* notation and an *interval* notation.

2. Find the coordinate (a, b) if a line passes through $(-2, 0)$ and (a, b) has gradient 2, and a line passes through $(3, 5)$ and (a, b) has gradient $3/4$.

3. Given two functions $f(x) = \sqrt{x^2 - 9}$ and $g(x) = \dfrac{1}{x}$.

   (a) Find $\left(\dfrac{f}{g}\right)(x)$ and its domain.

   (b) Find $(f \circ g)(x)$ and its domain.

4. Given $f(x) = 2 + \sqrt{25 - x^2}$ with $0 \leq x \leq 5$.

   (a) Find $f^{-1}(x)$ and its domain.

   (b) Sketch the graphs of $f(x)$ and $f^{-1}(x)$ in one coordinate system.

SOLUTIONS

1. **(25 points)** We split the compound absolute-value inequality $3 \le |2x-5| \le 7$ into two cases according to the sign of $2x-5$. **[5 points]**

   (a) $2x-5 \ge 0$ (i.e. $x \ge 5/2$): **[6 points]**

   $$3 \le 2x-5 \le 7 \implies 8 \le 2x \le 12 \implies 4 \le x \le 6.$$

   (b) $2x-5 < 0$ (i.e. $x < 5/2$): **[6 points]**
   The inequality becomes $3 \le -(2x-5) \le 7$, i.e. $3 \le 5-2x \le 7$:

   $$3 \le 5-2x \le 7 \implies -2 \le -2x \le 2 \implies -1 \le x \le 1.$$

   Taking the union of both cases, the set notation is: **[4 points]**

   $$\boxed{S = \{x \in \mathbb{R} \mid -1 \le x \le 1 \text{ or } 4 \le x \le 6\}.}$$

   In interval notation: **[4 points]**

   $$\boxed{S = [-1,\ 1] \cup [4,\ 6].}$$

2. **(25 points)** Let (a, b) be the unknown point.
   From the gradient conditions:

   (a) Line through $(-2, 0)$ and (a, b) has gradient 2: **[6 points]**

   $$\frac{b-0}{a-(-2)} = 2 \implies b = 2a+4. \tag{1}$$

   (b) Line through $(3, 5)$ and (a, b) has gradient $3/4$: **[6 points]**

   $$\frac{b-5}{a-3} = \frac{3}{4} \implies 4b = 3a+11. \tag{2}$$

   Substituting (1) into (2) and solving for a: **[6 points]**

   $$\begin{aligned} 4(2a+4) &= 3a+11, \\ 8a+16 &= 3a+11, \\ 5a &= -5, \\ a &= -1. \end{aligned}$$

   Substitute $a = -1$ into (1): **[4 points]**

   $$b = 2(-1)+4 = 2.$$

   Therefore, the required coordinate is: **[3 points]**

   $$\boxed{(a, b) = (-1,\ 2).}$$

   Verification:

   - Gradient $(-2, 0) \to (-1, 2)$: $\dfrac{2}{1} = 2$. ✓
   - Gradient $(3, 5) \to (-1, 2)$: $\dfrac{-3}{-4} = \dfrac{3}{4}$. ✓

3. **(25 points)** Given $f(x) = \sqrt{x^2 - 9}$ and $g(x) = \dfrac{1}{x}$.

Preliminary domains:

- $D_f = (-\infty, -3] \cup [3, +\infty)$ (need $x^2 - 9 \geq 0$).
- $D_g = \mathbb{R} \setminus \{0\}$ (need $x \neq 0$).

(a) **(12 points)** Compute the quotient: **[5 points]**

$$\left(\frac{f}{g}\right)(x) = \frac{\sqrt{x^2 - 9}}{\dfrac{1}{x}} = x\sqrt{x^2 - 9}.$$

The domain requires $x \in D_f \cap D_g$ and $g(x) \neq 0$. Here, $g(x) = 1/x$ is never zero. The remaining conditions are $x^2 - 9 \geq 0$ and $x \neq 0$. **[4 points]**
Since $x \leq -3$ or $x \geq 3$ already excludes $x = 0$: **[3 points]**

$$\boxed{D_{f/g} = (-\infty, -3] \cup [3, +\infty).}$$

(b) **(13 points)** Compute the composition: **[5 points]**

$$(f \circ g)(x) = f\left(\frac{1}{x}\right) = \sqrt{\frac{1}{x^2} - 9} = \frac{\sqrt{1 - 9x^2}}{|x|}.$$

Domain requires $x \neq 0$ and $g(x) \in D_f$, i.e. $\dfrac{1}{x} \leq -3$ or $\dfrac{1}{x} \geq 3$:

- $\dfrac{1}{x} \geq 3$: $\dfrac{1 - 3x}{x} \geq 0 \Rightarrow 0 < x \leq \dfrac{1}{3}$.
- $\dfrac{1}{x} \leq -3$: $\dfrac{1 + 3x}{x} \leq 0 \Rightarrow -\dfrac{1}{3} \leq x < 0$.

The domain conditions and their solution receive **[5 points]**.
Combining both intervals: **[3 points]**

$$\boxed{D_{f \circ g} = \left[-\tfrac{1}{3}, 0\right) \cup \left(0, \tfrac{1}{3}\right].}$$

4. **(25 points)** Given $f(x) = 2 + \sqrt{25 - x^2}$, $0 \leq x \leq 5$.

(a) **(13 points)** Let $y = 2 + \sqrt{25 - x^2}$. Isolate the square root and solve for x^2: **[4 points]**

$$\begin{aligned} y - 2 &= \sqrt{25 - x^2}, \\ (y - 2)^2 &= 25 - x^2, \\ x^2 &= 25 - (y - 2)^2. \end{aligned}$$

Take the nonnegative root because $x \geq 0$: **[3 points]**

$$x = \sqrt{25 - (y - 2)^2}.$$

Swapping $x \leftrightarrow y$: **[3 points]**

$$\boxed{f^{-1}(x) = \sqrt{25 - (x - 2)^2}.}$$

Domain of f^{-1} = Range of f. Since $f(0) = 7$, $f(5) = 2$, and f is strictly decreasing: $D_{f^{-1}} = [2, 7]$. **[3 points]**

(b) **(12 points)** Both curves are quarter-circle arcs (radius 5) symmetric about the line $y = x$.

Sketch criteria:

- Correct arc of f from $(0, 7)$ to $(5, 2)$. **[4 points]**
- Correct arc of f^{-1} from $(2, 5)$ to $(7, 0)$. **[4 points]**
- Reflection relationship about $y = x$ shown correctly. **[2 points]**
- Both curves share clearly labelled axes and a consistent scale. **[2 points]**

